Industrial Automation

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Industrial Automation (A12022)

Short Title:	Industrial Automation		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Automotive, Automation and IoT		
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Credits	5	**Level:**	Intermediate

Description of Module / Aims

The aim of this module is to introduce the student to network topologies, bus types and communication protocols. Scada topologies, PLC, electrical distribution, robotic and factory automation examples are developed by the student in addition to developing examples of Distributed Control Systems topologies, robotic and factory automation.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (`WD_EAUTO_B`)	2 / 4 / M
	BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	2 / 4 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	2 / 4 / M
	BEng (Hons) (`WD_ETRIC_B`)	2 / 4 / M
`ENGR-0093`	BEng in Electrical Engineering (`WD_EELCT_D`)	3 / 6 / M

Indicative Content

- Review of the control of actuators and sensors using PLCs and Labview.
- Scada Theory Overview
- Introduction seminar on Scada topologies, PLC, electrical distribution, robotic and factory automation examples.
- Develop Scada topologies, PLC, electrical distribution, robotic and factory automation examples.
- Distributed Control systems Theory Overview
- Introduction seminar on Distributed Control Systems topologies, electrical distribution, robotic, vision and factory automation examples.
- Develop Distributed Control Systems topologies, electrical distribution, robotic and factory automation examples.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Apply, interface and develop computer network topologies, bus types and communication protocols.
2. Integrate with Scada topologies, PLC, electrical distribution, robotic and factory automation examples.
3. Integrate with Distributed Control Systems topologies, electrical distribution, robotic and factory automation examples.
4. Demonstrate the ability to produce a practical automation system.
5. Demonstrate the professional, ethical and effective communication skills of a professional engineer.

Learning and Teaching Methods

- The course will consist of lectures/seminars in the laboratories together with extensive laboratories, mini project and formative assessments.
- The emphasis on course delivery will to engage the student in the learning of the material, through the integration of the course materials, with the real case studies in laboratories and mini project.
- The formative assignments will be used to provide an opportunity for the student to engage with the material, before it is applied, to maximise the learning opportunities for the student.
- Self-directed learning, moodle, and recommended software.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	50%		1,2,3
Continuous Assessment	50%		
Assignment		15%	2,4
In-Class Assessment		20%	4,5
Project		15%	1,4,5

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Medida, S. _Industrial Automation Pocket Guide_. .: IDC, 2003.

- Zhang, P. _Industrial Control Technology_. .: William Andrew, 2008.

Supplementary Material

- Buchli, J. _Mobile Robots, Moving Intelligence_. .: Advanced Robotic Systems International, 2006.
- Fuller, J.L. _Robotics: Introduction, Programming, and Projects_. 2nd ed.. New Jersey: Prentice Hall, 1999.
- Kurfess, T. _Robots and automation Handbook_. .: CRC Press, 2005.
- Niku, S.B. _Introduction to Robotics: Analysis, Systems, Applications_. .: Pearson Education, 2001.

Requested Resources

- ENGINEERING LAB: Electrical